

Operator Discovery Report: restart_controller_adaptive_2x_pool

- Operator type: restart_controller.
- Host scaffold: nearest_neighbor_2opt.
- Core idea: Deterministic restart controller that grows restart count moderately under stagnation, with a small seed pool and optional perturbation restarts to help escape long 2-opt plateaus on bottleneck/two-cluster instances.
- Novelty classification: adaptive_restart_controller.
- Rediscovery signature: restart_controller:2:7:2.
- Surviving candidate: False.
- Problem class where it helps: clustered_tsp, two_cluster_bottleneck_tsp.

Pseudocode

- Start from a conservative restart budget.
- Increase restart count only after the configured stagnation trigger.
- Keep perturbation restarts bounded by a deterministic maximum.

Complexity

- Non-empty lines: 12.
- Complexity score: 0.44.
- Fingerprint: 827019f4ff1e864b.

Why It Should Help

- The operator changes restart pressure under stagnation, which is a compact way to regulate exploration without rewriting search internals.

Failure Cases

- Does not transplant cleanly into sparse_three_opt.
- Runtime inflation is high at 0.623816.

Transplant Results

- nearest_neighbor_2opt gap delta 0.0, runtime delta 201.539986, distance-eval delta 5590.857143.
- cheapest_insertion_2opt gap delta -0.002995, runtime delta 844.364843, distance-eval delta 4462.857143.
- random_restart_2opt gap delta 0.0, runtime delta -171.145486, distance-eval delta -10716.0.
- sparse_three_opt gap delta 0.038805, runtime delta -4.093, distance-eval delta -367.857143.
- clustered_local_search gap delta 0.002851, runtime delta -127.175214, distance-eval delta -6300.285714.

Instance-Family Results

- heldout_tsplib operator gap 0.235058 vs baseline 0.235058.
- clustered_tsp operator gap 0.083393 vs baseline 0.110853.

- elongated_corridor_tsp operator gap 0.003204 vs baseline 0.003204.
- grid_like_tsp operator gap 0.106696 vs baseline 0.106696.
- nearest_neighbor_trap_tsp operator gap 0.0 vs baseline 0.0.
- two_cluster_bottleneck_tsp operator gap 0.105127 vs baseline 0.109057.
- uniform_euclidean operator gap 0.0 vs baseline 0.0.

Pareto Results

- Gap delta: -0.005462.
- Runtime inflation: 0.623816.
- Distance-evaluation inflation: 0.763082.
- Same-gap-faster flag: False.
- Better-gap-same-runtime flag: False.

Ablation Results

- disabled mean gap 0.121317 and delta vs full operator 0.005462.
- fixed_restart_budget mean gap 0.115856 and delta vs full operator 0.0.
- wider_restart_budget mean gap 0.115856 and delta vs full operator 0.0.